

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1157

Aurora 100 kV
October 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1157. The Interconnection Customer proposes a 207.8 MW solar photovoltaic facility interconnecting at the Aurora 100 kV substation in the Illinois area. The request was received on February 23, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$361,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
AURORA / WEST SPARTA 100 KV line	ELECTRIC LIGHT / AURORA 100 KV	101.7	109	69.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
AURORA SUBSTATION / W SPARTA 100 KV line	Reconductor line	\$431,000	84% (\$361,000)
	Total allocated to project		\$361,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.